

Python for data science

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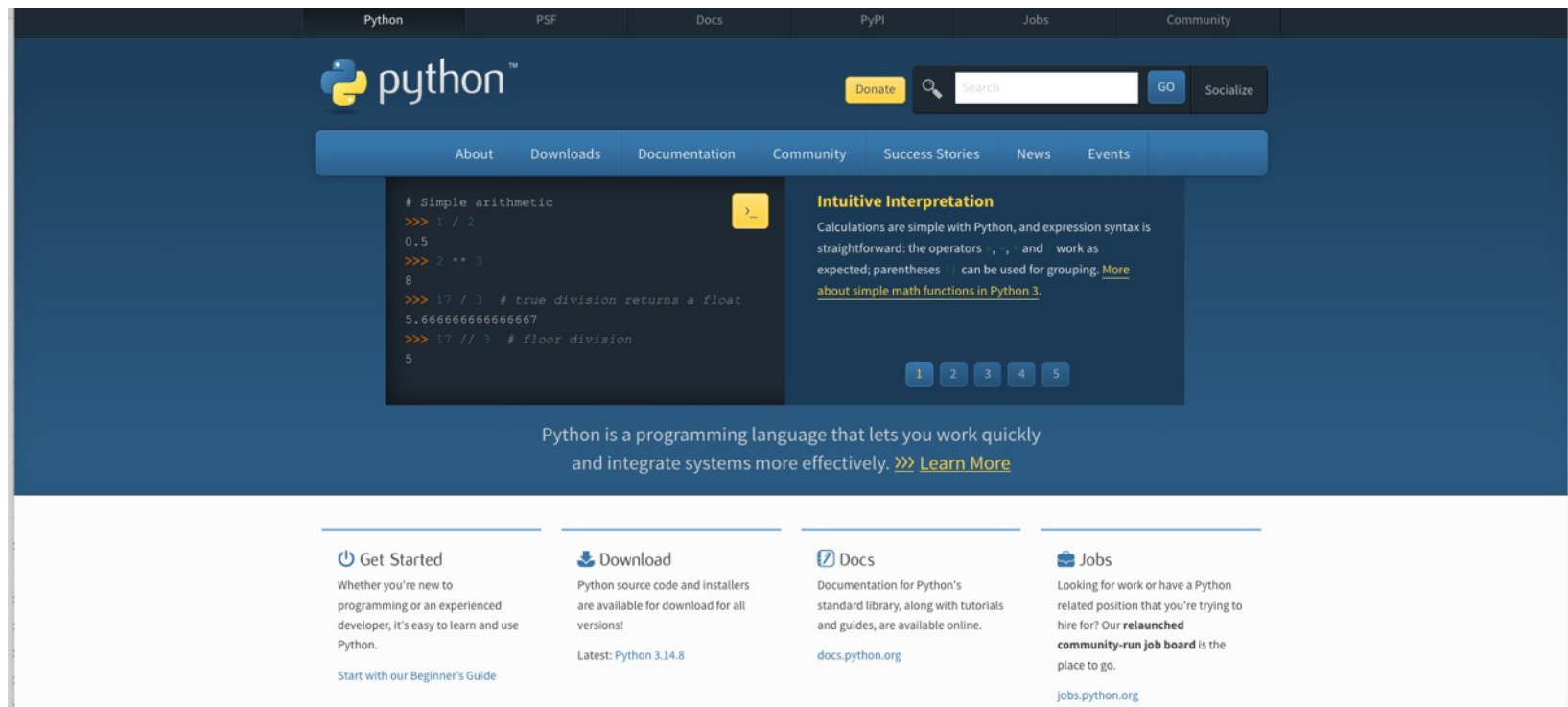
REQUIREMENTS PYTHON 101

INSTALL AN IDE + PYTHON

- IDE stands for *Integrated Development Environment*
Coding workspace
- Step 1 : Install VS CODE (open source)
<https://code.visualstudio.com/>
- Step 2 : Install a Python distribution
How to install Python from your browser ?
(see the next slides)
- Step 3 : Install Python extension in vs code (in order to be able to use it!)
(see the next slides)
- Step 4: Verify your installation by creating a small .py (python) script !
(see the next slides)

How to install Python from your browser ?

- Go to python.org



The screenshot shows the Python.org homepage with a dark blue header and a lighter blue main content area. The header includes navigation links for Python, PSF, Docs, PyPI, Jobs, and Community. A search bar and a 'Donate' button are also present. Below the header, a secondary navigation bar contains links for About, Downloads, Documentation, Community, Success Stories, News, and Events. The main content area features a code editor on the left with a simple arithmetic example, a 'Intuitive Interpretation' section on the right, and a footer with four columns: Get Started, Download, Docs, and Jobs. The code editor shows a simple arithmetic example with comments. The 'Intuitive Interpretation' section explains that calculations are simple with Python and provides a link to 'More about simple math functions in Python 3'. The footer columns provide links to get started, download Python source code and installers, access documentation, and find jobs.

python™

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```
# Simple arithmetic
>>> 1 / 2
0.5
>>> 2 ** 3
8
>>> 17 / 3 # true division returns a float
5.666666666666667
>>> 17 // 3 # floor division
5
```

Intuitive Interpretation

Calculations are simple with Python, and expression syntax is straightforward: the operators `/`, `*`, and `**` work as expected; parentheses `()` can be used for grouping. [More about simple math functions in Python 3.](#)

1 2 3 4 5

Python is a programming language that lets you work quickly and integrate systems more effectively. [>>> Learn More](#)

Get Started

Whether you're new to programming or an experienced developer, it's easy to learn and use Python.

[Start with our Beginner's Guide](#)

Download

Python source code and installers are available for download for all versions!

Latest: Python 3.14.8

Docs

Documentation for Python's standard library, along with tutorials and guides, are available online.

docs.python.org

Jobs

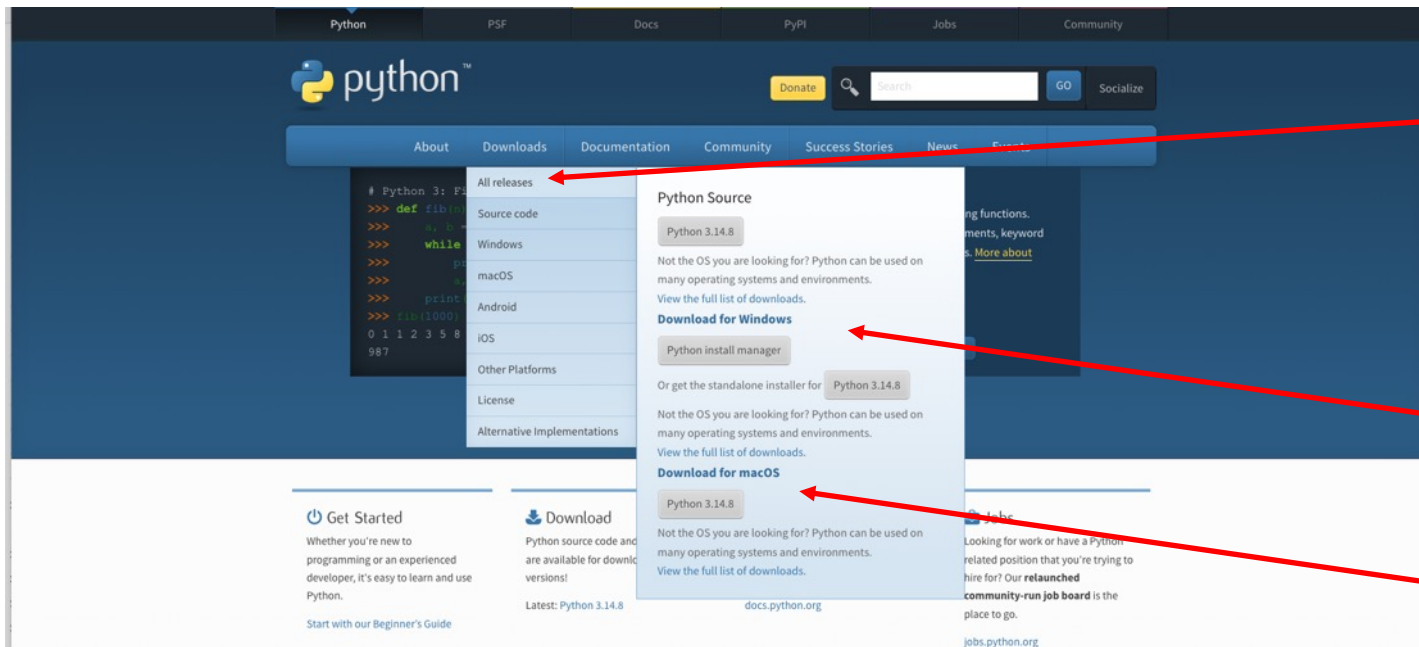
Looking for work or have a Python related position that you're trying to hire for? Our **relaunched community-run job board** is the place to go.

jobs.python.org

How to install Python from your browser ?

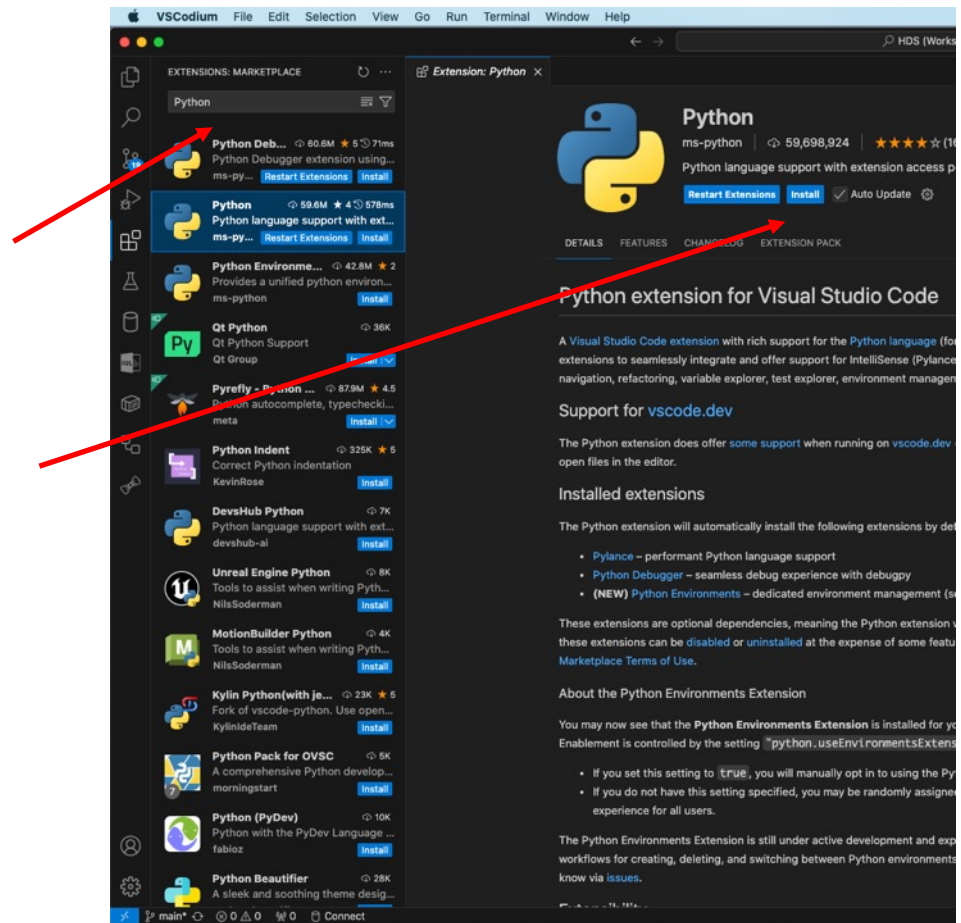
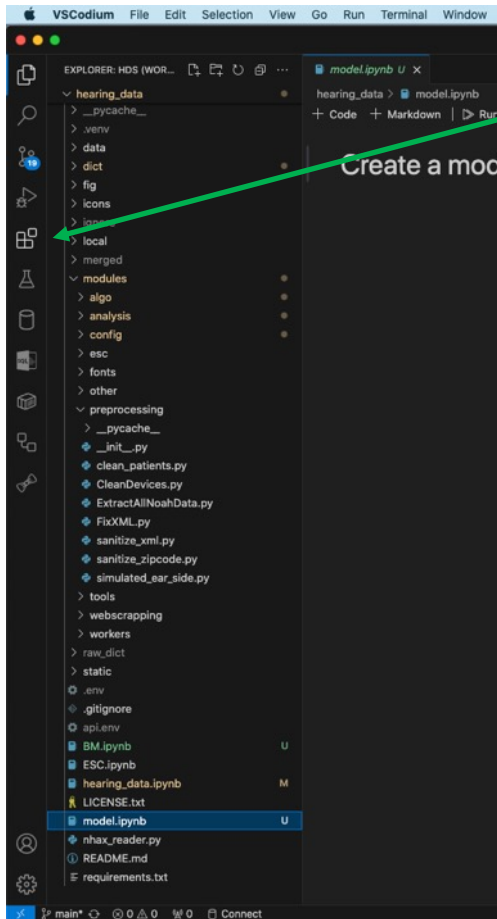
Go to Downloads button, then All releases, and click on Python 3 depending on your operating system.

WINDOWS ONLY : Make sure to check the box that says "Add python.exe to PATH" at the bottom of the installation window **before** clicking Install Now.



Install Python extension in vs code ?

In vs code, go to the **extension icon**, type **Python** (only) and install it !
You're all set! Just need to control the installation (next slide) !



Note : VS Codium and VS Code have the same GUI. I recommend VS Code.

Verify your installation!

- In Vs code click on the new file icon and create it and name it : `hello.py`
- In the file type : `print("Hello world!")` and save the file.
- Click the play button to run your script. If you see `Hello world!` in your terminal, your environment is ready! See you on campus.
- Don't be afraid we will review each definition together.



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Troubleshooting : Python not found

1. Windows only!
2. Close VS Code !
3. Run the Python installer (.exe) file again.
4. In the advanced options, check "Add Python to environment variables" (or PATH).
5. Click install/modify
6. Reopen VS Code and you're all set 😊